



What is Wrong with These Surveys?



Identify what is wrong in each of these surveys. Be sure to explain.

1. The mayor of Springfield is interested in finding out the average age of people in the city. He obtains a list of all the homeowners in the city and then contacts a simple random sample of 300 people. He uses the sample to estimate the average age of all the people in the city.

a. What is wrong with this survey?

b. Do you think the mayor will over or underestimate the true mean age of people in Springfield? Why?

2. The principal at a school wants to know the proportion of students who did all their homework last night. She selects a simple random sample of 100 students and send an email to each of them asking if they did all their homework last night. Of the 40 responses, 36 of the students said that they did all their homework last night (90%).

a. What is wrong with this survey?

b. Do you think the administration will over or underestimate the true proportion of students who did all their homework last night? Why?

3. There is a proposal to cut some of the funding for the fire department in a city. One of the firefighters in the city wants to know the proportion of people in his neighborhood who support the budget cuts. He takes a random sample of 30 homes and visits them dressed in his uniform.

a. What is wrong with this survey?

b. Do you think the firefighter will over or underestimate the true proportion of his neighbors who support the budget cuts? Why?

Lesson 2.4 – Potential Problems with Sampling

QuickNotes

Check Your Understanding

1. A counselor at a large high school wants to learn about student opinion regarding remote learning versus in-person learning. Each of the following sampling methods introduces a possible source of bias. Name and explain the type of bias.
 - a. The counselor selects a random sample of students from those who signed up for remote learning.
 - b. The counselor selects a random sample of 100 students and sends them all an email, but only 28 students respond.
 - c. The counselor surveys the first 50 students that submit their learning preference for the upcoming school year.

2. Students at a certain high school completed a survey that included the following question:

Social media use can negatively affect teens, distracting them, disrupting their sleep, and exposing them to bullying, rumor spreading, unrealistic views of other people's lives and peer pressure. How many hours per week do you use social media?

What type of bias is present? Will the survey underestimate or overestimate the true average number of hours per week that students use social media? Explain.